

AERO62520 Coursework-3 Design Requirements Analysis

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1 Problem Statement

1.1 Custom Requirement Statement

"Develop a robot which can autonomously retrieve coloured objects from the environment and place them in matching storage bins located at the starting point."

1.2 Problem Statement

Growing labour shortages have many industries seeking to automate simple tasks. Autonomous navigation, object recognition and manipulation are tasks that span the field of robotics. A robotic system that fulfils these requirements will have many uses in the wider world and serve as a good example of robotic capability. Primarily designed as a learning exercise this project provides an opportunity to explore different robotic systems and their interplay whilst also attempting to mimic a real-world collaborative industry scenario.

2 Concept of Operations (CONOPS)

2.1 CONOPS State Transitions

This state-machine diagram Fig. 1 provides a compact mission logic overview, linking all CONOPS phases and the decision gates that move the rover forward or trigger recovery. It answers: (1) Where in the mission loop are we? (2) What condition must be satisfied to advance? (3) What happens on failure or operator intervention?

2.2 Start and Setup

Purpose: As shown in Fig. 2, the operator places the rover in the designated Start Area, confirms arena readiness (walls, bins, declared colours, and static obstacle placement), powers sensors and actuators, and configures declared run parameters. Operators confirm sensor health and mission parameters on the operator interface and issue the start command to begin autonomous operation.

Preconditions: Start Area placement by operator and availability of declared run parameters.

Postconditions: Rover is powered, sensors and actuators pass health checks, run parameters are recorded, and the mission manager reports READY.

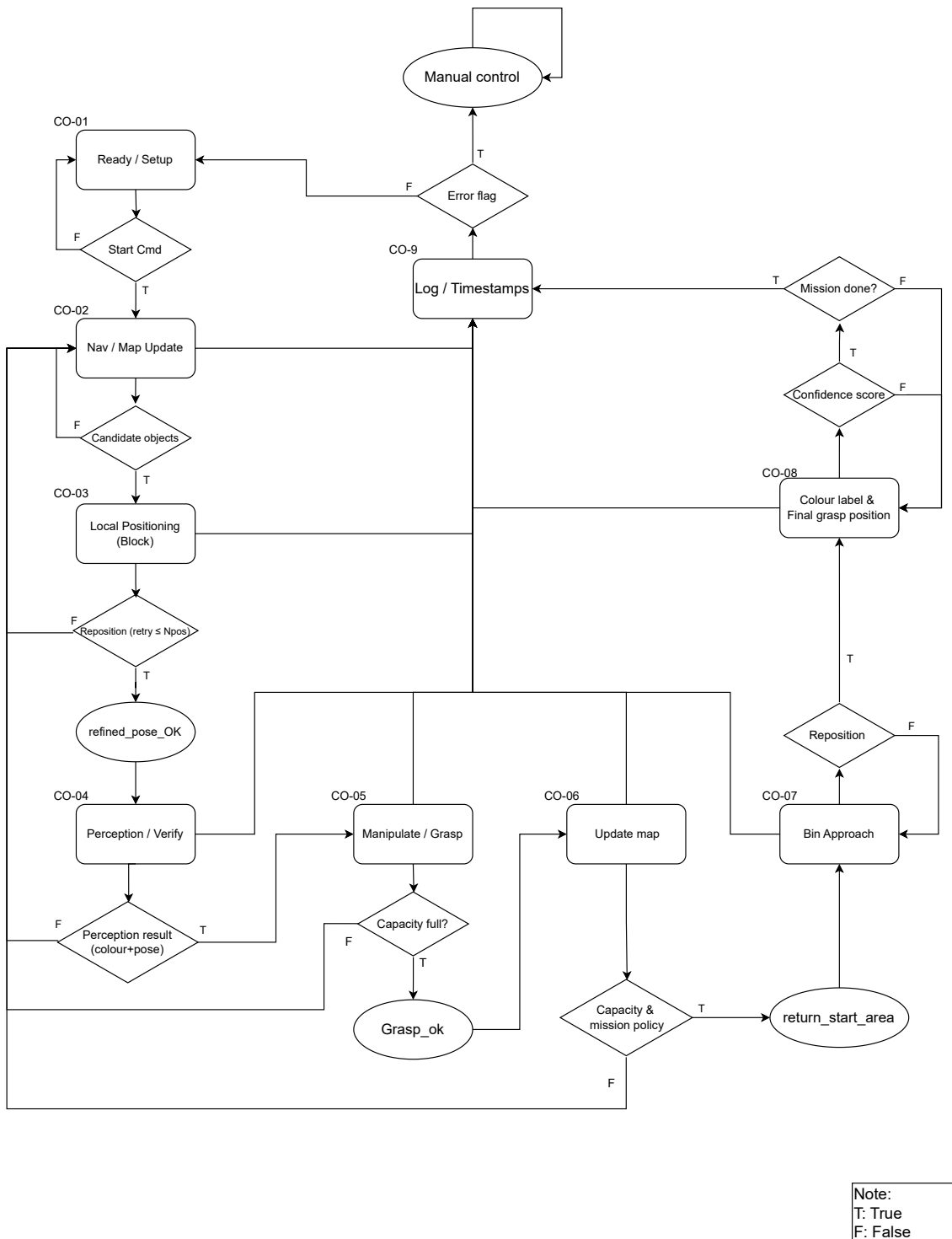


Fig. 1. State-machine for CONOPS: shows phase flow from CO-01 READY through CO-02 mapping, CO-03 local positioning, CO-04 to CO-06 grasping and map update, CO-07 to CO-09 deposit, plus error handling, timestamps and manual control.

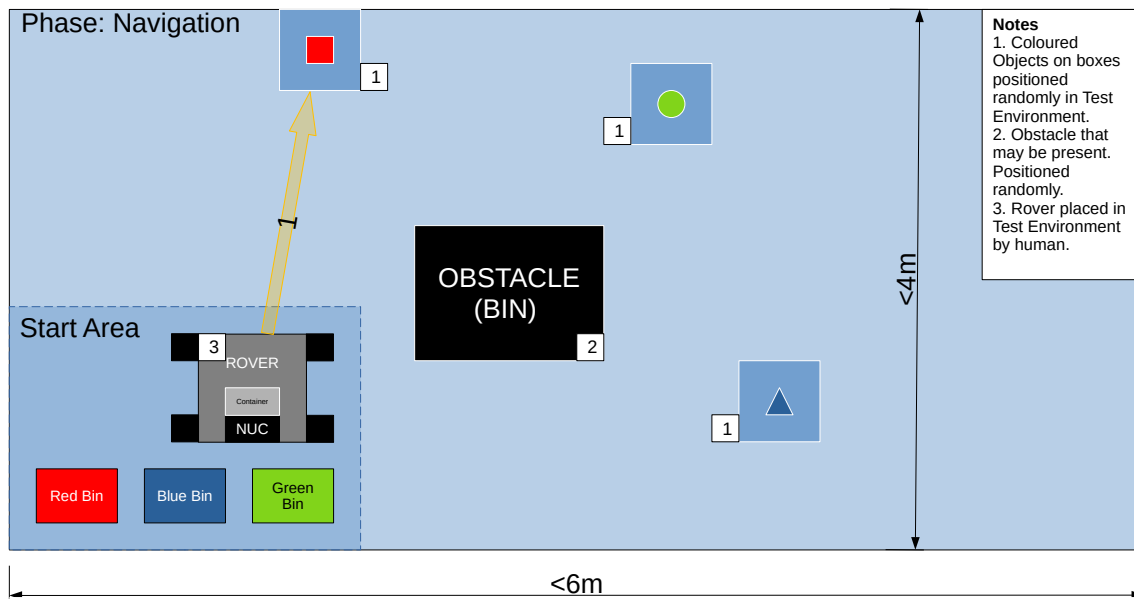


Fig. 2. CO-01 Navigation setup: rover positioned in Start Area beside colour-coded bins; planned route (yellow arrow) toward scattered target zones is shown and a large black rectangle marks a static obstacle.

Recovery: If any critical sensor or actuator fails health checks, the system alerts the operator and allows sensor restart or manual intervention; if recovery fails within the allowed setup window the run is aborted.

2.3 Autonomous Navigation to Search

Purpose: As shown in Fig. 3, after start, the rover departs the Start Area and performs arena-wide autonomous traversal to search for coloured blocks using onboard LIDAR and camera sensors while respecting arena boundaries and avoiding static obstacles.

Preconditions: LIDAR point clouds, camera images, map/boundary data, and declared colour set.

Postconditions: Candidate object detections with coarse positions and an updated local occupancy map.

Recovery: Upon localization drift or odometry failure, switch to visual re-localization or return to the Start Area for manual recovery.

2.4 Local Positioning at Target

Purpose: As shown in Fig. 4, on detection of a candidate block or bin, the rover executes a local approach routine to position itself adjacent to the target using fine depth/LIDAR alignment to reach a stable, collision-safe pose suitable for perception and manipulation.

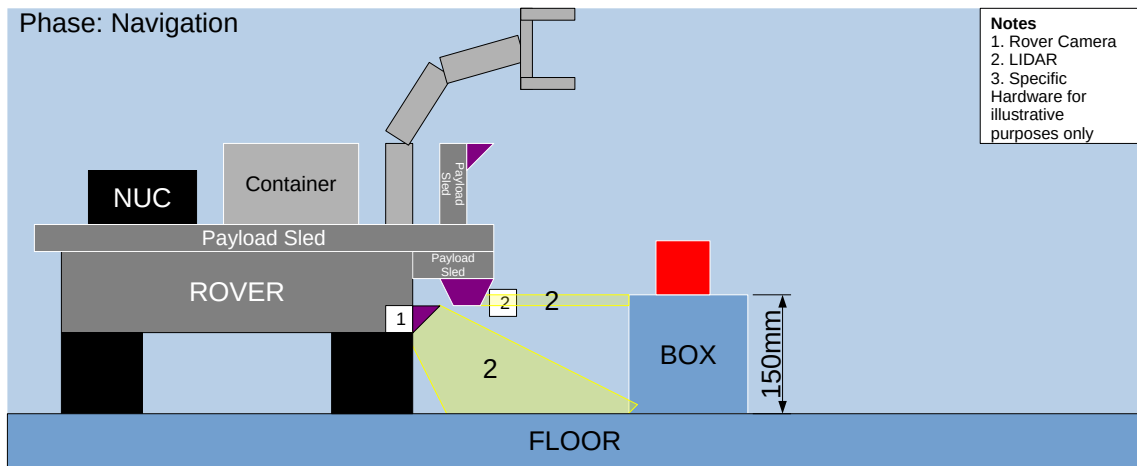


Fig. 3. Navigation-phase diagram for CO-02: the rover uses camera and LIDAR (sensor cones shown) to scan ahead, detect a red coloured object on a box, and update the local occupancy map while avoiding obstacles.

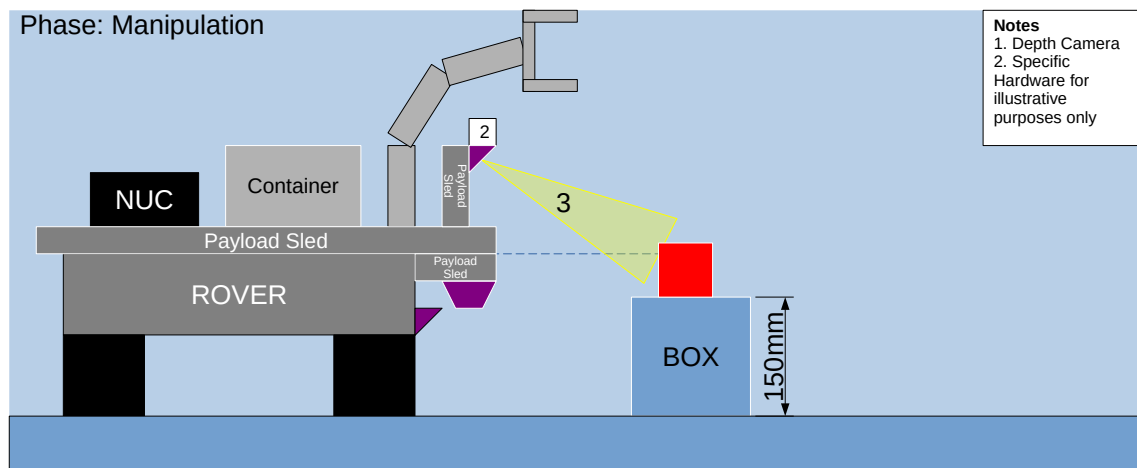


Fig. 4. CO-03 Local positioning: the rover aligns using depth camera and LIDAR (sensor cones) to refine the block pose and attain a collision-safe manipulator approach (300 mm clearance shown)

Preconditions: Coarse candidate pose, vehicle odometry, and close-range sensor data.

Postconditions: A refined target pose relative to the robot and a verified safe manipulator approach pose.

Recovery: If the rover cannot achieve a collision-free grasping pose within N retries, the target is marked as unavailable, and the rover resumes its search.

2.5 Perception and Verification

Purpose: As shown in Fig. 5, sensors refine object localization, classify candidate objects against the declared colour set, and publish pose plus confidence for downstream decision-making. Perception must confirm the object meets classification and pose-confidence thresholds before manipulation is attempted.

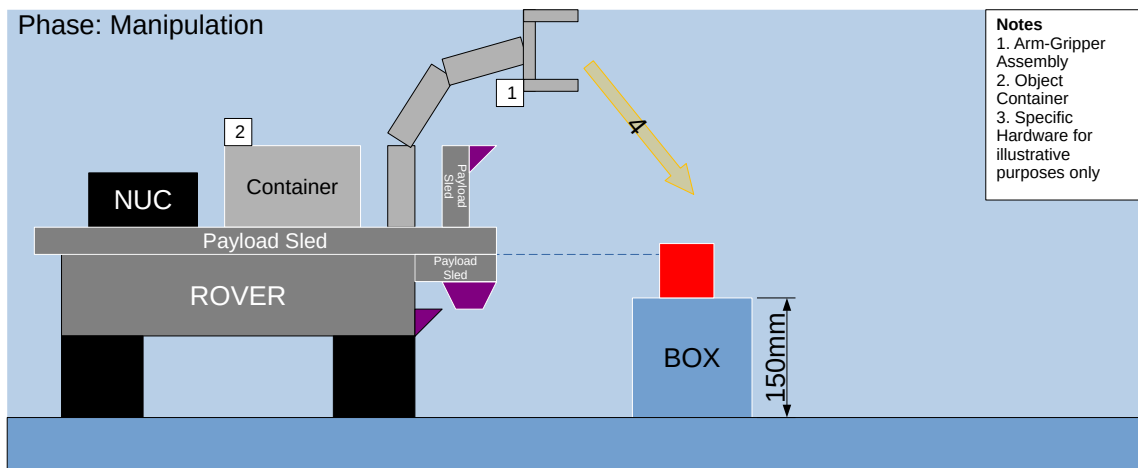


Fig. 5. CO-04 Perception: sensors refine block pose and colour classification; confidence metric confirms readiness for manipulation.

Preconditions: Sensor colour data, candidate pose and declared colour labels.

Postconditions: Final manipulation pose, colour label, and confidence score.

Recovery: On low confidence, system classifies as false positive and continues search.

2.6 Manipulation

Purpose: As shown in Fig. 6, the manipulator aligns the end effector to the final manipulation pose, executes end-effector sequence, verifies success with onboard sensors (force, tactile, or visual), and secures the object for transport. If verification fails, recovery behaviours are executed.

Preconditions: Final manipulation pose and end-effector parameters from perception.

Postconditions: End-effector success/failure flag and object deposited in payload location.

Recovery: On end-effector failure perform safe retreat, adjust approach or end-effector parameters, and retry up to a configured attempt limit, if still unsuccessful mark target as failed.

2.7 Robot Returns to Start Area

Purpose: As shown in Fig. 7, based on mission policy (e.g., capacity reached or ordered mode rules), the rover returns to the Start Area or repositions for further retrieval/ deposition tasks.

Preconditions: Storage/capacity state, current map, Start Area pose, mission policy.

Postconditions: Rover pose returns to Start Area or prepared deposition location, mission state update (arrived), and deposition request if applicable.

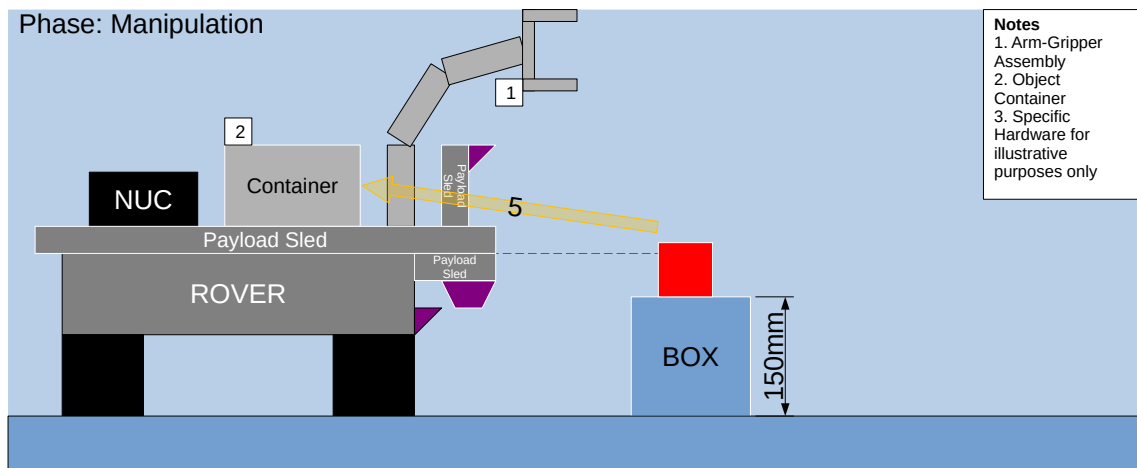


Fig. 6. CO-05 Manipulation: the arm aligns to the final pose and follows the indicated path to grasp the red block on the 150 mm box, then places it into the payload container; success is verified by onboard sensing.

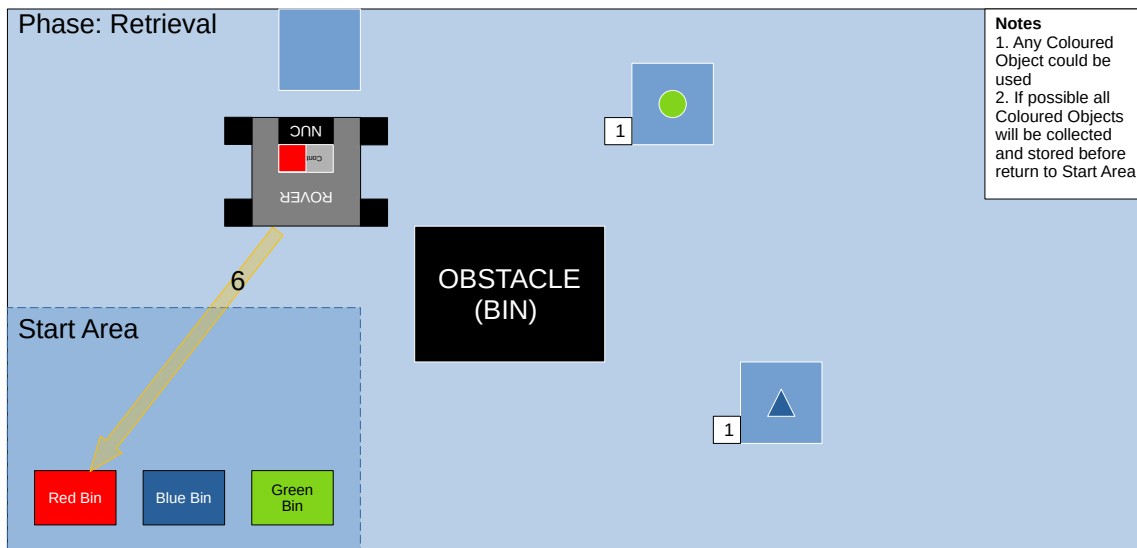


Fig. 7. CO-06 Return to Start Area: the rover follows the indicated path to the Start Area and colour bins while avoiding the central obstacle.

Recovery: If path blocked, re-plan or choose alternate route; if localization fails, perform visual re localization or request operator assist.

2.8 Robot Positions Adjacent to Coloured Bin

Purpose: The robot uses sensor guidance for navigation and positioning itself beside the target- Coloured Bin to establish a safe approach path for Coloured Object delivery, as shown in Fig. 8.

Preconditions: Bin coarse position, sensor data, odometry.

Postconditions: Approach pose for bin deposition (robot base pose), local map update.

Recovery: If bin not found or occluded, attempt alternate viewpoints or use operator input.

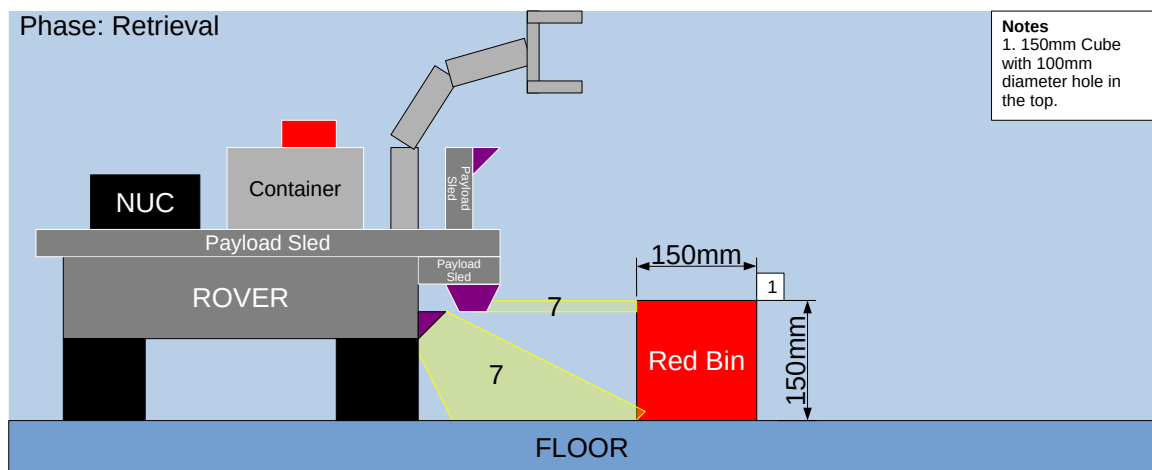


Fig. 8. CO-07 Bin adjacency: the rover halts with a 150 mm standoff to the coloured bin and aligns a clearance-safe approach corridor (shaded) for deposition.

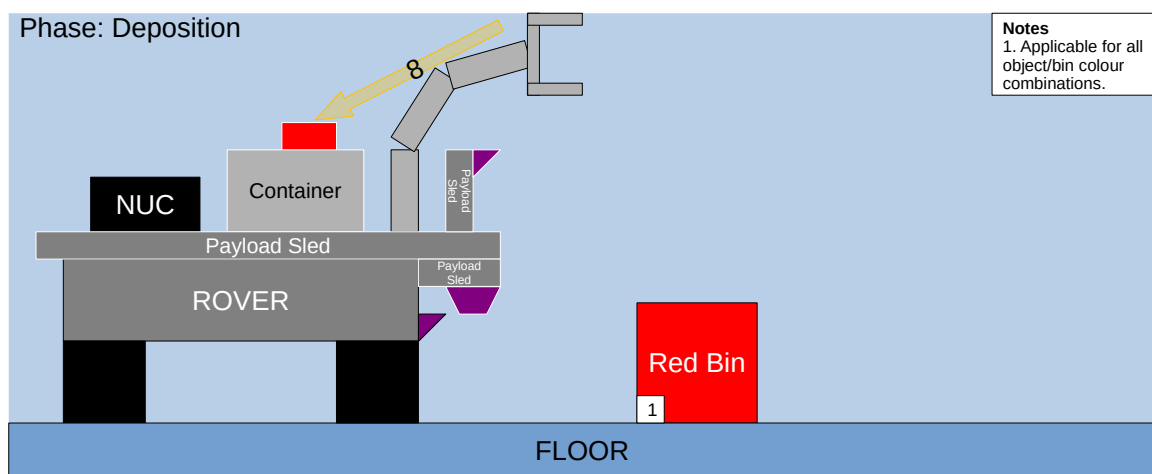


Fig. 9. CO-08 Deposition: rover verifies the red bin, lifts the block from the payload sled/container, and initiates a clearance-safe swing to transfer the block toward the matching floor bin.

2.9 Deposition into Colour-Matching Bin

Purpose: As shown in Fig. 9, the rover approaches the declared bin on the floor, verifies bin pose and colour, aligns the manipulator with safe clearance from the payload sled/container, and deposits the block; ordered-mode constraints are enforced if active, shown in Fig. 10.

Inputs: Confirmed bin pose and colour, block pose in gripper, deposition plan/payload item info.

Outputs: Deposition success/failure record, updated payload inventory and task counts.

Recovery: On misplacement, drop, or bounce-out, reposition and retry up to the allowed attempts; on persistent failure, log the event and either continue the mission or request operator intervention according to mission rules.

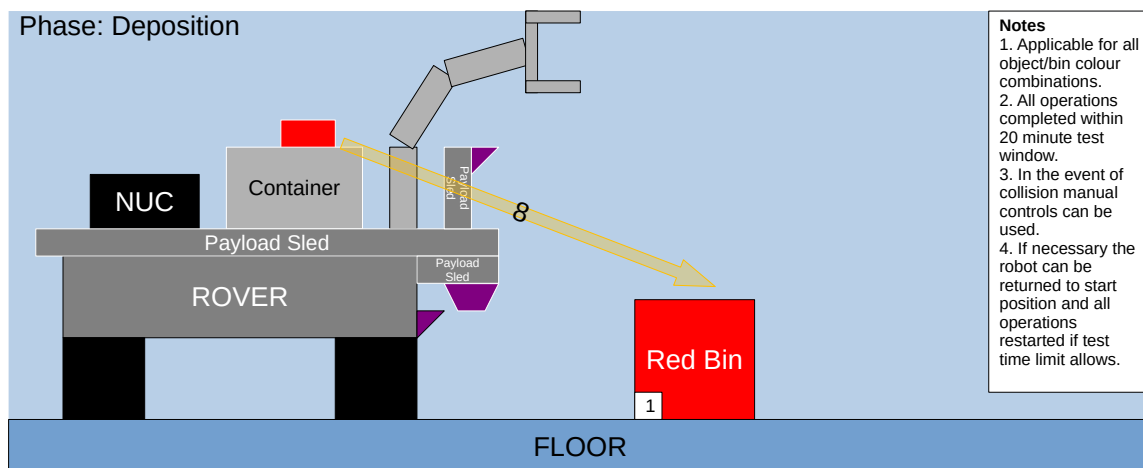


Fig. 10. CO-09 Deposit: the arm follows the indicated path and releases the block into the matching red bin on the floor; success is confirmed and logged, with retry on miss per mission rules.

2.10 Safety, Operator Controls, and Mission End

Purpose: Define safety controls and operator interfaces: users may pause, emergency-stop, resume, or restart via hardware and operator interfaces. In case of collision or unrecoverable fault the system permits manual intervention and restart within the test window. At mission end the system performs safe shutdown and archives logs.

Preconditions: Emergency/pause signals, fault detection, and mission-timer status.

Postconditions: System safe state, archived logs, and restart/terminate decision.

Requirements: Emergency-stop must place drive systems into a safe state and log the event with timestamps.

3 Functional Requirements

Table 1. Functional Requirements

Require- ment ID	Requirement	Source
Operation		
FR-01	The robot shall operate within the Test Environment.	Customer requirements Statement; Stakeholder engagement
FR-01.1	The boundary will be defined by a white wall < 0.5 m high.	Stakeholder engagement
FR-01.2	The Test Environment will be 4 m < x < 6 m.	Collected Data; Stakeholder engagement
FR-01.3	The Test Environment will be level.	Stakeholder engagement
FR-02	The robot shall be capable of restarting operation.	Engineering Judgement
FR-02.1	The robot shall achieve full operation in <5 minutes.	Stakeholder engagement
FR-03	All hardware shall be mounted on the robot platform.	Stakeholder engagement
FR-03.1	Combined weight of hardware shall be less than 5 kg.	Data sheet - Leo Rover 1.8
Navigation and Planning		
FR-04	The robot shall navigate autonomously.	Customer requirements Statement; Stakeholder engagement
FR-04.1	The robot shall avoid obstacles and unintended collisions.	Engineering Judgement
FR-04.2	No external controls shall be used except in the event of a collision or unexpected behaviour.	Stakeholder engagement
FR-04.3	Path planning should avoid tangling of cables (if present).	Risk Assessment
FR-04.4	Path planning shall locally generate paths for navigation from sensor data.	Stakeholder engagement

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Table 1 – continued

Require- ment No.	Requirement	Source
FR-04.5	Maximum linear velocity shall be limited to ≤ 250 mm/s.	Risk Assessment
FR-04.6	Maximum angular velocity shall be $\leq 60^\circ/\text{s}$.	Data sheet - Leo Rover 1.8
Vision and Perception		
FR-05	The robot shall locate & classify the Coloured Objects.	Stakeholder engagement; Engineering Judgement
FR-05.1	The colour of the Coloured Objects will be declared by Team 12.	Stakeholder engagement
FR-05.2	The minimum size of the Coloured Objects shall be 20 x 20 x 20 mm.	Collected Data
FR-05.3	The robot shall locate & classify at a distance >75 mm.	Engineering Judgement
FR-05.4	The robot perception shall tolerate lighting variation including shadows and differing shades of the same nominal colour.	Stakeholder engagement; Engineering Judgement
FR-06	The robot shall locate & classify the Coloured Bins.	Stakeholder engagement
FR-06.1	The Coloured Bins will be 150 x 150 x 150 mm.	Stakeholder engagement
FR-06.2	The robot perception shall tolerate lighting variation including shadows and differing shades of the same nominal colour.	Stakeholder engagement; Engineering Judgement
Object Manipulation		
FR-07	The robot shall manipulate the Coloured Objects.	Stakeholder engagement
FR-07.1	The shape of the Coloured Objects will be specified by Team 12.	Stakeholder engagement
FR-07.2	The minimum size of the Coloured Objects shall be 20 x 20 x 20 mm.	Collected Data
FR-07.3	The robot shall manipulate at a height of 150 mm.	Engineering Judgement
FR-07.4	The maximum weight shall be 150 g.	Data sheet - myCobot 280pi
FR-07.5	The positional accuracy shall be ± 1 mm.	Data sheet - myCobot 280pi
FR-08	The robot shall transport the Coloured Objects.	Stakeholder engagement
FR-08.1	The robot shall transport a minimum of 1 Coloured Object.	Stakeholder engagement
FR-08.2	The robot should transport a maximum of 3 Coloured Objects.	Stakeholder engagement
FR-09	The robot shall deposit the Coloured Objects into the corresponding Coloured Bins.	Stakeholder engagement

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Table 1 – continued

Require- ment No.	Requirement	Source
FR-09.1	The robot shall deposit through a 100 mm Diameter Hole in the top of the Coloured Bin.	Stakeholder engagement
Safety		
FR-10	The robot shall be capable of being manually handled.	Risk Assessment
FR-10.1	The robot will have a mass <20 kg.	Risk Assessment
FR-10.2	The robot should have a form factor that allows proper lifting technique.	Risk Assessment
FR-10.3	The robot should not have any payloads which could cause injury.	Risk Assessment
FR-11	The robot shall be able to be stopped in the case of unexpected behaviour.	Risk Assessment
FR-11.1	The robot shall be capable of being issued a stop command wirelessly.	Risk Assessment
FR-11.2	The robot shall have a form of easily accessible emergency shutdown button or switch.	Risk Assessment
Power		
FR-12	The robot should operate without trailing cables.	Stakeholder engagement
FR-12.1	Onboard power should provide for a minimum of 25 minutes.	Engineering Judgement
FR-12.2	Electrical design and maintenance shall comply with lab risk assessment.	Stakeholder engagement

4 Performance Requirements

Table 2. Performance Requirements

Require- ment No.	Performance Requirement	Source of Functional Requirement / Limitation
PR-01	The system shall traverse the arena using onboard sensors without reliance on a preloaded map.	Stakeholder Engagement 1 & 2; Engineering Judgement

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Table 2 – continued

Require- ment ID	Performance Requirement	Source of Functional Requirement / Limitation
PR-02	The rover shall achieve a nominal linear speed of 0.4 m/s and angular speed of 60°/s for effective traversal. Traversal speed shall remain within 0.12–0.4 m/s, with fine approach speed ≤ 0.12 m/s for depth-assisted alignment.	Stakeholder Engagement 1 & 2; Engineering Judgement; Manufacturer data sheet — Leo Rover 1.8; Intel RealSense D435i
PR-03	The system shall support arena dimensions published prior to testing and complete setup within 5 minutes.	Stakeholder Engagement 2; Engineering Judgement
PR-04	The system shall support sequential retrieval of blocks; multi-grasp is optional. Block size shall be between 20 mm and 75 mm.	Stakeholder Engagement 2; Engineering Judgement
PR-05	The system shall deposit blocks into colour-matched bins.	myCobot280pi; myCobot280 Adaptive Gripper; Stakeholder Engagement 1 & 2; Engineering Judgement
PR-06	The system should support ordered collection mode when declared before run. Blocks shall be deposited in declared order.	Stakeholder Engagement 2; Engineering Judgement
PR-07	The navigation stack shall detect and avoid static obstacles.	(Source implied from navigation requirements) Stakeholder Engagement 2; Engineering Judgement
PR-08	The system shall reject non-target-coloured objects.	Stakeholder Engagement 2; Engineering Judgement
PR-09	The system shall operate in a flat arena with no elevation changes or ramps.	Stakeholder Engagement 2; Engineering Judgement
PR-10	The system shall allow restart during testing; restart time shall count against mission time.	Stakeholder Engagement 2; Engineering Judgement
PR-11	The system shall operate untethered on battery for ≥ 20 minutes unless an approved alternative is used.	Leo Rover 1.8; Stakeholder Engagement 2; Engineering Judgement
PR-12	The gripper shall support declared jaw opening and payload capacity for standard block geometry. Gripping range shall be 20–45 mm; block weight 150 g; payload ≤ 250 g.	myCobot280 Adaptive Gripper; myCobot 280pi; Engineering Judgement
PR-13	The system shall sample at ≥ 5 Hz for obstacle detection.	SLAMTEC RPLIDAR A2M12; Engineering Judgement
PR-14	The system shall operate only on the declared colour set submitted before the run.	Stakeholder Engagement 2; Engineering Judgement
PR-15	The system shall publish object poses with ≤ 50 mm position error and $\leq 10^\circ$ orientation error.	Intel RealSense D435i; myCobot 280pi; Engineering Judgement
PR-16	Calibration shall be performed during development only; no runtime calibration is required.	Stakeholder Engagement 1 & 2; Engineering Judgement
PR-17	The planner shall generate reactive paths using onboard sensor data only; no global map shall be used. Replan latency shall be ≤ 500 ms.	Engineering Judgement
PR-18	Pose estimates shall be updated at ≥ 10 Hz using odometry and sensor data with timestamp sync error ≤ 10 ms.	Leo Rover 1.8; ASUS NUC12WSK; Engineering Judgement

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Table 2 – continued

Require- ment ID	Performance Requirement	Source of Functional Requirement / Limitation
PR-19	Navigation status, current goal, and completion events shall be published with ≤ 200 ms latency.	Engineering Judgement
PR-20	Emergency stop shall halt actuators and enter safe state within 200 ms; hardware override supported.	Leo Rover 1.8; Engineering Judgement
PR-21	Battery voltage and current shall be logged at ≥ 1 Hz and alarms published when thresholds are crossed.	Leo Rover 1.8; Engineering Judgement
PR-22	A consistent TF tree shall be published at ≥ 10 Hz with correct parent-child relations and time sync.	myCobot 280pi; Intel RealSense D435i; Engineering Judgement
PR-23	TF documentation should include frame hierarchy, coordinate conventions, and time sync approach.	Engineering Judgement
PR-24	On-field setup shall complete within 5 minutes excluding development calibration.	Stakeholder Engagement 2; Engineering Judgement
PR-25	Mission execution shall complete within 20 minutes; pack-up within 5 minutes; restarts counted.	Stakeholder Engagement 1 & 2; Engineering Judgement
PR-26	Perception shall publish a colour classification confidence value and require confidence ≥ 0.80 before permitting manipulation actions.	Engineering Judgement; Intel RealSense D435i
PR-27	Manipulator pose acceptance before grasping shall be lateral error ≤ 10 mm and angular error $\leq 10^\circ$.	Engineering Judgement; Manufacturer data sheet — myCobot 280pi; Intel RealSense D435i
PR-28	Minimum safety standoff from walls/obstacles shall be ≥ 0.25 m.	Engineering Judgement; Manufacturer data sheet — SLAMTEC RPLIDAR A2M12

5 Requirements Verification Matrix

Table 3. Performance Requirements Verification Matrix

Require- ment No.	Shall Statement	Verification method	Verification of success criteria	Responsible Party	Results
1	The robot shall operate within the Test Environment.	Demonstration	Robot shown operation is not impeded by any factors of the testing environment.	All	TBC

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Table 3 – continued

Require- ment No.	Shall Statement	Verification method	Verification of success criteria	Responsible Party	Results
2	The robot shall be capable of restarting the operation.	Demonstration	Robot can be shown to restart the operation after interruption/from off state.	AB	TBC
2.1	The robot shall achieve full operation in <5 minutes.	Test	In multiple trials the robot can be restarted and fully operational within the 5 minute window.	AB, ZP	TBC
3	All hardware shall be mounted on the robot platform.	Demonstration	There is no un-mounted hardware.	AB, ZP	TBC
3.1	Combined weight of payload shall be less than 5kg.	Analysis	The combined calculated weight of the payload is less than 5kg.	NL, AB	TBC
4	The robot shall navigate autonomously.	Test	The robot repeatedly navigates the environment without external influence for an extended period of time.	SH, QS	TBC
4.1	The robot shall avoid obstacles and unintended collisions.	Demonstration	During operation the robot does not impact the test environment wall, deliberate obstacles, coloured bins or boxes unintentionally.	SH, AB	TBC
4.2	No external controls shall be used except in the event of a collision or unexpected behaviour.	Demonstration	The robot navigates autonomously unless the specified situations occur.	SH	TBC
4.4	Path planning shall locally generate paths for navigation from sensor data.	Test	The robot successfully navigates random environments without preloaded data.	SH	TBC

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Table 3 – continued

Require- ment No.	Shall Statement	Verification method	Verification of success criteria	Responsible Party	Results
4.5	Maximum linear velocity shall be limited to ≤ 250 mm/s.	Test	The robot does not exceed 250 mm/s during test operation.	SH	TBC
4.6	Maximum angular velocity shall be $\leq 60^\circ$ /s.	Test	The robot does not exceed 60° /s during test operation.	SH	TBC
5	The robot shall locate classify the Coloured Objects.	Test	The robot provides positional data (within tolerance) and correctly identifies the object's colour.	QS, AB	TBC
5.2	The minimum size of the Coloured Objects shall be 20 x 20 x 20 mm.	Test	The robot shall correctly locate and classify blocks that measure 20 mm on a side.	QS, AB	TBC
5.3	The robot shall locate classify at a distance >75 mm.	Test	The robot provides accurate positional data and correctly identifies the object's colour whilst positioned adjacent to the coloured object's mounting box.	QS, AB	TBC
5.4	The robot perception shall tolerate lighting variation including shadows and differing shades of the same nominal colour.	Demonstration	The robot can be shown to correctly identify an object's colour under the natural laboratory lighting conditions.	QS, AB	TBC
6	The robot shall locate classify the Coloured Bins.	Test	The robot provides accurate positional data and correctly identifies the bins' colour during testing.	QS, AB	TBC
6.2	The robot perception shall tolerate lighting variation including shadows and differing shades of the same nominal colour.	Demonstration	The robot can be shown to correctly identify the bins' colour under the natural laboratory lighting conditions.	QS, AB	TBC

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Table 3 – continued

Require- ment No.	Shall Statement	Verification method	Verification of success criteria	Responsible Party	Results
7	The robot shall manipulate the Coloured Objects.	Test	The robot shall successfully and repeatedly manipulate the Coloured Objects in the desired way.	NL, ZP	TBC
7.2	The minimum size of the Coloured Objects shall be 20 x 20 x 20 mm.	Test	The robot shall successfully manipulate objects no smaller than 20 mm on a side.	NL, ZP	TBC
7.3	The robot shall manipulate at a height of 150 mm.	Demonstration	The task space of the robot shall include 150 mm above the ground with margin.	NL, ZP	TBC
7.4	The maximum weight shall be 150 g.	Analysis	When calculated, none of the coloured blocks shall weigh more than 150 g.	NL, ZP	TBC
7.5	The positional accuracy shall be ± 1 mm.	Demonstration	The robot shall demonstrate that it can complete manipulation tasks that require precision.	NL, ZP	TBC
8	The robot shall transport the Coloured Objects.	Test	The robot will repeatedly carry coloured objects between given locations.	NL, ZP, SH	TBC
8.1	The robot shall transport a minimum of 1 Coloured Object.	Demonstration	The robot will show it can transport a single coloured object inside the test environment.	NL, ZP, SH	TBC
9	The robot shall deposit the Coloured Objects into the corresponding Coloured Bins.	Test	The robot will show that it can repeatedly deposit the Coloured object into the correct bin.	NL, ZP	TBC

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Table 3 – continued

Require- ment No.	Shall Statement	Verification method	Verification of success criteria	Responsible Party	Results
9.1	The robot shall deposit through a 100 mm Diameter Hole in the top of the Coloured Bin.	Demonstration	The accuracy of the manipulation will be sufficient to show the coloured objects being deposited through the hole.	NL, ZP	TBC
10	The robot shall be capable of being manually handled.	Demonstration	The robot shall be picked up and placed into the test environment.	NL, ZP	TBC
10.1	The combined weight of the robot and the payload shall be <20 kg.	Analysis	When calculated the sum of weights of all the components and the payload shall be less than 20 kg.	NL, AB	TBC
11	The robot shall be able to be stopped in the case of unexpected behaviour.	Demonstration	Emergency shutdown shall be demonstrated.	ZP	TBC
11.1	The robot shall be capable of being issued a stop command wirelessly.	Demonstration	The robot will show the ability to be stopped remotely.	SH, QS	TBC
11.2	The robot shall have a form of easily accessible emergency shutdown button or switch.	Inspection	Emergency stop button is present, labelled, and accessible.	ZP, SH	TBC
12.2	Electrical design and maintenance shall comply with lab risk assessment.	Demonstration	The power supply and batteries shall comply with the risk assessment by design and when inspected.	AB, ZP	TBC